

PARALI SE PAISA

Satellite AI that turns paddy straw into verified farmer income

Concept Note | Greenovators Hackathon 2026 | Primary Theme: Waste to Wealth
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1. Background

Every October and November, farmers in Punjab and Haryana get only a short window between the paddy harvest and wheat sowing to clear their fields. Punjab alone has about 30 lakh hectares under paddy this year and expects around 18.81 million tonnes of straw [1]. Haryana produces about 7.3 million tonnes every year [2].

This straw is no longer just waste. Industrial boilers, pellet plants, CBG plants and biomass power plants used an estimated 7.06 million tonnes of Punjab's straw in 2025 [3], and Haryana pays farmers Rs 1,200 per acre to manage straw without burning it [2]. Even so, a large part of the straw is still burnt every season.

2. Problem Statement

In our view, the straw keeps getting burnt because of two gaps.

(a) No one can see the supply in time. Farmers, baler operators and buyers do not know which fields have been harvested, how much straw is lying where, or which fields are about to be burnt. Balers reach late, buyers are unsure of supply, and the farmer, who has only a few days to sow wheat, burns the straw.

(b) Burning is not being measured properly anymore. Official numbers count active fires seen by satellites that pass over between about 10:30 am and 1:30 pm, and farmers have moved burning to the late afternoon and evening [5]. More than 90% of farm fires in Punjab in 2024 and 2025 happened after 3 pm, compared to only 3% in 2021 [4]. Punjab's official fire count fell from 36,663 in 2023 to 5,114 in 2025 [1], but the burnt area seen in Sentinel-2 images fell much more slowly and was still about 20,000 sq km in 2025 [5]. Haryana releases its incentive only after field inspection [2], which is slow when 5.65 lakh farmers have registered 39.33 lakh acres [6].

So money and machines do not reach the right field at the right time, and there is no reliable proof of who did not burn.

3. Proposed Solution

Parali se Paisa is a satellite and AI based platform that follows every paddy field during the harvest season and helps turn its straw into income before it is burnt. It has four parts:

1. **Field status monitoring:** every few days, a machine learning model classifies each field from free Sentinel-2 images as standing crop, harvested (straw lying), burnt or sown. Sentinel-1 radar, which can see through cloud and smoke, is used as a cross-check.
2. **Straw supply and burn-risk prediction:** for each harvested field, we estimate the straw available and predict the chance that it will be burnt in the next few days.
3. **Smart baler dispatch:** high-risk fields are assigned to the nearest free balers. Baling is planned for dry days, since industries need straw moisture below 20% [7], and bales are sent to the nearest buyer that has demand.
4. **No-burn certificate:** each registered field gets a certificate with before and after satellite images and dates. It can be used to release the incentive faster, and it gives buyers a record of where their straw came from.

Farmers use the system through a simple chatbot in Punjabi and Hindi. Officials, baler operators and buyers use a web dashboard. An example message to a farmer (translated):

“Your field looks harvested. A baler can come on Thursday morning. After pickup you will get your no-burn certificate.”

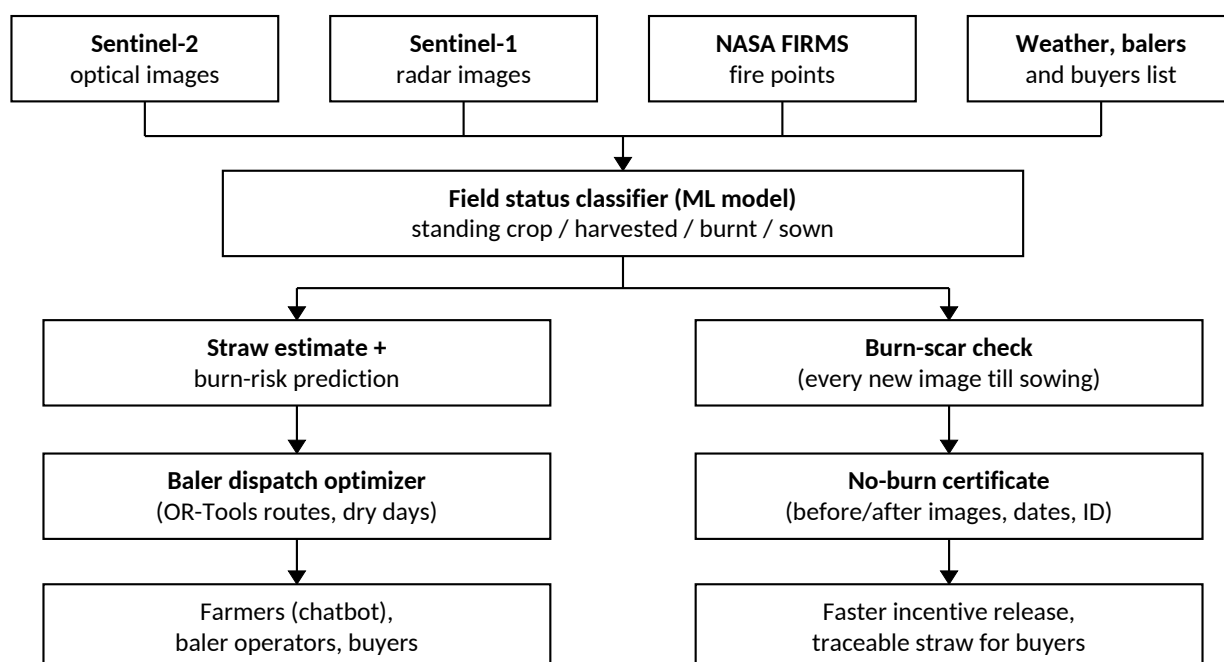


Figure 1: Overall working of Parali se Paisa

4. How It Works

4.1 Data sources

- **Sentinel-2 (Level-2A):** 10 to 20 m resolution, a new image about every 5 days (free).
- **Sentinel-1 radar:** 10 m, sees through cloud and smoke (free).
- **NASA FIRMS fire points (VIIRS, 375 m):** to confirm burnt patches (free).
- **Weather forecast:** rain and humidity, for planning dry-day baling.
- **Balers and buyers list:** sample data for the pilot, later from state CRM records.

4.2 Detecting field status

For each 10 m pixel we calculate two standard indices,

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4}) \quad \text{NBR} = (\text{B8A} - \text{B12}) / (\text{B8A} + \text{B12})$$

along with BAIS2, a burned area index made for Sentinel-2 [8], and Sentinel-1 backscatter. A Random Forest / LightGBM model trained on labelled pixels from the 2023 to 2025 seasons gives one of the four classes. Labels come from clear before/after image pairs, and FIRMS points help confirm burnt patches. Pixels are then grouped by field.

Table 1: Field status and the signal we look for

Field status	What the image shows	Main signal
Standing crop	Green, healthy vegetation	High NDVI
Harvested, straw lying	Sudden loss of green, dry straw on top	NDVI drops, no burn signal
Burnt	Dark, charred surface	NBR falls sharply, BAIS2 rises
Sown (wheat)	Bare soil, then new green growth	Low NDVI, then rising

4.3 Straw estimate and burn-risk model

Straw available = harvested area $\times$ average straw yield. Punjab's figures (18.81 million tonnes from about 30 lakh ha) give roughly 6 tonnes per hectare [1], which we adjust using the crop's peak NDVI. A gradient boosting model then predicts the chance that a harvested field is burnt within 3 days, using days since harvest, how close the wheat sowing deadline is, past burning in that area, distance to the nearest baler and buyer, and the rain forecast.

4.4 Baler dispatch

We use Google OR-Tools (vehicle routing) to assign balers to fields each day, aiming to clear the most straw from high-risk fields. Constraints are baler capacity (roughly one hour per acre, with about 12 to 15 quintals of bales per acre [9]), working hours, travel distance, buyer demand and dry weather.

4.5 No-burn certificate

For each registered field, the system checks every new image from harvest until sowing. If no burn signal appears, it generates a PDF certificate with dates, before/after images and a unique ID that officials can verify online. Unclear cases are flagged for field inspection instead of being approved automatically.

5. What Is New in Our Approach

- **Burn scars, not fire counts.** A fire can be timed to avoid the satellite pass, but the burnt surface still shows up in later Sentinel-2 images [10]. We verify each field by burnt area, whatever time the fire was lit.
- **Predict and prevent.** We act in the gap when a field is harvested but not yet burnt, with a pickup offer and payment instead of a penalty.
- **One data backbone for everyone.** The same satellite data tells baler operators where the straw is, tells industries what supply is coming, gives officials automatic verification and helps farmers get paid faster.
- **No hardware and very low cost.** Only free satellite data and open-source tools, so it can be extended to Uttar Pradesh and Madhya Pradesh, where fire counts are rising [5].

6. Hackathon Plan

Pilot district: Sri Muktsar Sahib, Punjab (about 2.03 lakh ha under paddy and basmati [11]). As the hackathon falls in the harvest season, the demo will use real satellite images from the same week.

Table 2: Day-wise build plan

Day	Work	Output
Day 1 (7 Oct)	Earth Engine pipeline for Sentinel-2 and Sentinel-1; training and testing the field status classifier	Field status map of the pilot district
Day 2 (8 Oct)	Straw estimate, burn-risk model, OR-Tools dispatch, web dashboard	Daily baler plan shown on a map
Day 3 (9 Oct)	Chatbot (Punjabi/Hindi), no-burn certificate, testing on this week's images, final demo	Working end-to-end prototype

7. Expected Impact

Income: Biomass power plants in Punjab have paid about Rs 169 per quintal for paddy straw [11]. So every 1 lakh tonnes of straw that goes to buyers instead of being burnt is worth about Rs 17 crore to farmers and baler operators.

Pilot estimate for Muktsar: the district's straw was estimated at about 12.1 lakh tonnes [11]. If our system helps send just 5% more of it to buyers (about 60,500 tonnes):

- Income: $60,500 \text{ t} \times \text{Rs } 1,690 \text{ per tonne} \approx \text{Rs } 10.2 \text{ crore}$
- CO₂ avoided: $60,500 \text{ t} \times 1,460 \text{ kg per tonne} \approx 88,000 \text{ tonnes}$ [12]
- Particulate matter avoided: $60,500 \text{ t} \times 3 \text{ kg per tonne} \approx 180 \text{ tonnes}$ [12]
- Effort needed: about 45,000 acres of baling, i.e. roughly 225 balers for 20 days at 10 acres a day, which is exactly what our dispatch module plans.

Government: Haryana sanctioned Rs 471.96 crore of incentives in 2025 [6]. Satellite checks can cover every registered field every few days, so inspection teams only need to visit doubtful cases.

Industry: most companies surveyed by CEEW lose 15 to 25% of stored bales every year because of quality loss [7]. Dry-day baling and better supply forecasts can reduce this.

8. Limitations and How We Handle Them

- **Small fields:** many plots are small compared to 10 m pixels, so we report at field level only where a field has enough pixels, otherwise at village-block level.
- **Smoke and cloud:** Sentinel-1 radar fills the gap when optical images are blocked.
- **Quick ploughing after burning can hide the scar:** frequent revisits reduce this gap, and doubtful fields go for field inspection rather than automatic approval.
- **Field boundaries:** for now farmers mark their field on a map in the chatbot; later this can be linked to state land records.

9. Future Scope

Linking with state CRM portals and machine booking apps, adding wheat straw and other residues, expanding to Uttar Pradesh and Madhya Pradesh, and certified straw records for industries' ESG reporting.

References

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